

vides a wide and diverse field in electronics industry, either directly or through ancillary producers, explains Vivek Madhukar, COO of Times Business Solutions, which operates TimesJobs.com. He says, "The biggest employers are in IT, manufacturing/engineering, healthcare/biotechnology and auto sectors based in and around Bengaluru, Pune and Delhi. Salaries start at the lower end, but hands-on experience definitely improves one's growth prospects, with earnings reaching over 1.5 million rupees with adequate exposure."

Amit Bhargava, managing director, Asha Electronics believes the key to boost manufacturing sector lies with fresh graduates. He says, "Engineers need to realise that the real fun is in creating, that is, manufacturing. Young graduates can fire up the industry with their energy and ideas. Entering the realms of purchase, production, quality and marketing will energise these streams." He adds, "As production gets complex so will marketing, and this will help young graduates pick up lucrative assignments and incentives from companies. Starting as trainees, and setting themselves up in fields they excel in, will boost their growth prospects."

"Most of our designs are SMT-based and need high skill levels in assembly. Being a design house, we do hand-soldering as much as possible and depend highly on practical experience," says Anand. He adds, "At Knewron we welcome engineers from all strata; however, each one needs to demonstrate his worth before coming on board." Talking about his company, Bhargava informs, "Engineers are as such encouraged in our organisation to take up jobs in all functions. The field being complex, engineers are best suited for it."

Centum Electronics recruits as well as encourages fresher engineers to work on their state-of-the-art SMT lines. Srinivasa Bharadwaj, manager-engineering, Centum Electronics says, "As the SMT machines involve understanding of electronics assembly processes as well as maintenance of the pick-and-place equipment, where a lot of mechanical movement of the parts is involved, we prefer to have engineers from electronics and mechanical streams. Engineers from electrical stream also fit in."

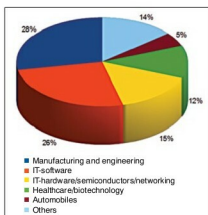


Fig. 1: Top industries hiring professionals for electronics manufacturing

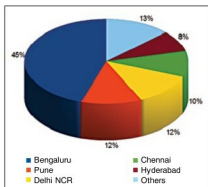


Fig. 2: Top locations for jobs in electronics manufacturing

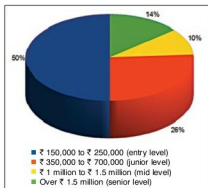


Fig. 3: Salary break-up for engineers in electronics manufacturing segment

Why a career in SMT

Fresh engineers interested in learning equipment functionality and process engineering are suitable for SMT lines. In addition, engineers with an aptitude of equipment maintenance can also take up the job of an SMT engineer. Singh says, "As it is the core of an electronics manufacturing industry, there is a lot of focus on the technological developments in this area. This makes it more interesting

and challenging for any of the fresh engineers to take up SMT as a career option."

According to Singh, SMT forms the back bone of electronics assembly line and is therefore a very important area for an electronics manufacturing plant. He says, "Though there is a requirement of experienced engineering resources running SMT assembly lines, it is very important to have a significant pool of fresh engineers working on SMT lines too." To that Bharadwaj adds, "Engineering graduates with a good aptitude for learning basic manufacturing processes like screen printing, solder-paste inspection, pick and place, reflow soldering and automatic inspections are preferred. It is very challenging for a fresh electronics engineer to work on fully-automatic computer-controlled equipment. The industry offers various roles, which include responsibilities like process engineering and equipment maintenance of the SMT lines."

What an SMT engineer learns

Talking about the exposure for an SMT engineer, Singh shares, "Since SMT assembly is the core of electronic assembly, an SMT engineer gets exposed to critical process areas like tooling design (such as stencil, palettes), pick-and-place programming, reflow profiling, AOI programming and x-ray inspection." Shifting to the equipment side, he adds, "Here, the areas where an engineer would get an insight into are breakdown maintenance, equipment effectiveness, equipment capabilities and new equipment selection, to name a few."

Learning about statistical process controls for critical processes like solder-paste printing, for instance, using very-high-end inspection equipment is another interesting area in SMT. "SMT first-pass yield is critical in electronic assembly. Hence, invariably, the SMT engineers have to learn about the process to bring in yield improvements. Therefore there is opportunity to learn and use statistical techniques to bring forth yield improvements," informs Bharadwaj.

Aptitude, right attitude and soft skills

According to Anand, for SMT-assembly and manufacturing-based jobs, people must have practical experience in soldering. He informs, "Even a fresher should